Table S1. Sensitivity of the worst-image gain to the budget fraction ρ.

All ten conditions are reported. Δ is proportional minus uniform, in percentage points; the pre-registered thresholds were A $\geq$ 10, B $\geq$ 5, C $< 5. N = 548 images; ρ = 0.5 was pre-registered.

head	ρ	uniform worst (%)	proportional worst (%)	Δ (pp)	Δ micro (pp)	verdict
one-to-many	0.30	6.940	16.667	+9.727	0.771	B
one-to-many	0.40	9.148	19.355	+10.207	1.166	A
one-to-many	0.50	11.672	20.000	+8.328	1.380	B
one-to-many	0.60	14.826	20.000	+5.174	1.334	B
one-to-many	0.70	17.981	23.333	+5.352	1.063	B
one-to-one	0.30	6.940	16.129	+9.189	0.738	B
one-to-one	0.40	8.833	16.667	+7.834	1.241	B
one-to-one	0.50	11.356	16.667	+5.310	1.401	B
one-to-one	0.60	14.196	22.222	+8.027	1.329	B
one-to-one	0.70	17.974	22.222	+4.248	1.133	C

Table S2. A deployable variant of the proportional allocation.

All three allocations use the same total budget ($\rho = 0.5$), the same integer rounding and the same top-up, so the only difference is the allocation signal. "oracle" uses the annotated object count n_i, unavailable at inference; "prediction" uses the number of predictions the detector emits at the largest budget, available before quotas are fixed. Δ is against the uniform allocation, in percentage points. $N = 548$ images.

case	p_i median	n_i median	Δworst oracle	Δworst prediction	Δmacro oracle	Δmacro prediction
s42	548	65	+8.33	+5.72	-6.36	-3.44
s42-o2o	1000	65	+5.31	+0.32	-6.14	-1.30
s43	533	65	+8.38	+6.55	-6.57	-3.49
s44	518	65	+8.01	+4.24	-6.87	-3.83

Table S3. The worst image on test-dev under both allocations.

Budget fraction $\rho = 0.5$. The same image, index 69, is the worst under both allocations and on both heads. Its annotated count n_i is 8, so its capacity is 8 boxes; the uniform quota already covers that capacity, and neither allocation recovers a single object. The two quota vectors nevertheless differ on most images, and both allocations spend the same total. $N = 1610$ test-dev images.

head	worst image	n_i	p_i	quota uniform	quota prop.	TP	recall	images with a different quota	spend
o2m	#69	8	113	8	4	0	0.0%	1568/1610	37551
o2o	#69	8	191	8	4	0	0.0%	1569/1610	37551

Table S4. Holm step-down correction, condition by condition.

The family is the 10 conditions of the ρ sweep (five fractions × two heads). One-sided null: the worst-image gain is not positive. Conditions are ordered by rank; the Holm threshold at rank k is $\alpha/(m - k + 1)$, and p_{adj} is the step-down adjusted value. N = 548 images, 10,000 bootstrap resamples, RNG seed 20261014. All 10 conditions remain rejected after correction.

rank	condition	Δ (pt)	p raw	Holm threshold	p adjusted	rejected
1	o2m/rho=0.30	+9.727	0.0001	0.0050	0.0010	yes
2	o2m/rho=0.40	+10.207	0.0001	0.0056	0.0010	yes
3	o2m/rho=0.50	+8.328	0.0001	0.0063	0.0010	yes
4	o2o/rho=0.30	+9.189	0.0001	0.0071	0.0010	yes
5	o2o/rho=0.40	+7.834	0.0001	0.0083	0.0010	yes
6	o2o/rho=0.60	+8.027	0.0001	0.0100	0.0010	yes
7	o2o/rho=0.50	+5.310	0.0034	0.0125	0.0136	yes
8	o2m/rho=0.60	+5.174	0.0038	0.0167	0.0136	yes
9	o2m/rho=0.70	+5.352	0.0050	0.0250	0.0136	yes
10	o2o/rho=0.70	+4.248	0.0054	0.0500	0.0136	yes

Table S5. Sensitivity of the main results to the label-count cap.

The main text caps every quota at $\min(n_i, p_i)$. Here the n_i term is removed, so quotas are capped at p_i alone and spend is the number of boxes actually emitted; true positives are still bounded by n_i. Every gain grows when the cap is removed, so the reported main results are conservative estimates for the label-count-capped convention. VisDrone2019-DET validation, seed 42, N = 548. Values are percentage points. Δ is proportional minus uniform.

head	ρ	Δ worst, capped	Δ worst, no cap	change	Δ micro, capped	Δ micro, no cap
o2m	0.30	+9.73	+10.36	+0.63	+1.28	+3.13
o2m	0.40	+10.21	+11.15	+0.95	+1.55	+4.04
o2m	0.50	+8.33	+9.59	+1.26	+1.57	+4.78
o2m	0.60	+5.17	+7.70	+2.52	+1.52	+5.55
o2m	0.70	+5.35	+8.51	+3.15	+1.26	+5.74
o2o	0.30	+9.19	+9.82	+0.63	+1.24	+3.05
o2o	0.40	+7.83	+8.46	+0.63	+1.61	+4.03
o2o	0.50	+5.31	+6.89	+1.58	+1.56	+4.80
o2o	0.60	+8.03	+10.23	+2.21	+1.51	+5.53
o2o	0.70	+4.25	+8.03	+3.78	+1.36	+5.63

Table S6. Worst-image gain under four top-up rules, three of which never read labels.

Every condition is the same without the oracle top-up: with the budget matched, the two label-free rules land within 0.32 points of the oracle arm. The no-top-up arm under-spends by up to 2.75% at the smallest budget fraction and is not a valid control there. VisDrone2019-DET validation, seed 42, N = 548. Δ worst-image recall, proportional minus uniform, in percentage points.

head	ρ	no top-up	largest remainder	most capacity	oracle	spend err %
o2m	0.30	+10.04	+10.04	+9.73	+9.73	0.006
o2m	0.40	+10.21	+10.21	+9.89	+10.21	0.004
o2m	0.50	+8.33	+8.33	+8.01	+8.33	0.003
o2m	0.60	+5.17	+8.51	+4.86	+5.17	0.002
o2m	0.70	+5.35	+5.35	+5.04	+5.35	0.001
o2o	0.30	+9.50	+9.50	+9.19	+9.19	0.006
o2o	0.40	+7.83	+7.83	+7.52	+7.83	0.004
o2o	0.50	+5.31	+5.31	+5.00	+5.31	0.003
o2o	0.60	+8.03	+8.03	+7.71	+8.03	0.002
o2o	0.70	+4.25	+4.25	+3.93	+4.25	0.001

Table S7. The gain sits in the lower tail; the median pays for it.

Change in order statistics of per-image recall, proportional minus uniform, in percentage points. Every condition has the same sign at the minimum, the 5th, 10th and 25th percentiles, and the opposite sign at the median, so the effect is a shift of the lower tail rather than a change confined to one image. VisDrone2019-DET validation, seed 42, N = 548.

head	ρ	Δ min	Δ p5	Δ p10	Δ p25	Δ median	worst image same?
o2m	0.30	+9.73	+9.68	+7.70	+4.26	-3.40	no
o2m	0.40	+10.21	+9.01	+7.63	+4.09	-2.90	no
o2m	0.50	+8.33	+8.65	+6.77	+2.97	-3.50	no
o2m	0.60	+5.17	+6.52	+5.00	+1.18	-2.31	no
o2m	0.70	+5.35	+5.84	+3.62	+0.64	-2.36	no
o2o	0.30	+9.19	+9.10	+7.26	+3.94	-3.01	no
o2o	0.40	+7.83	+8.74	+6.55	+4.05	-3.08	no
o2o	0.50	+5.31	+7.62	+6.03	+2.65	-3.29	no
o2o	0.60	+8.03	+7.17	+4.76	+1.51	-2.95	no
o2o	0.70	+4.25	+5.89	+2.92	+0.41	-2.64	no

Table S8. Output-budget utilisation across benchmark splits

The last three columns fix B = 100. ceiling is Σ_i min(B,n_i)/Σ_i n_i, the largest fraction of all annotated objects that an average per-image budget of B can in principle recover.

Values are exactly as computed; see Methods for definitions.

Split	Domain	Images (with gt)	mean n_i	max n_i	ceiling	util	frac(>100)
COCO val2017	ground	4,952	7.43	63	1.000	0.074	0.000
COCO train2017	ground	117,266	7.33	93	1.000	0.073	0.000
CrowdHuman train	ground	15,000	23.53	391	0.957	0.225	0.021
VisDrone test-dev	aerial	1,610	46.65	461	0.896	0.418	0.079
VisDrone val	aerial	548	70.73	317	0.890	0.630	0.195

Table S9. Worst-image gain from proportional allocation as a function of the budget

VisDrone2019-DET validation, N = 548. S = B·N is the total budget. Of the 17 swept budgets, the twelve with B ≤ 70 are spendable (spend/S = 1.00000); this table lists ten of them plus B = 100 and B = 300, where the budget can no longer be spent in full. util is the label-side utilisation from Table S8. Gains are in points; each Δ column lists seeds 42, 43, 44, and Δ micro is on the same allocation as the gain.

Values are exactly as computed; see Methods for definitions.

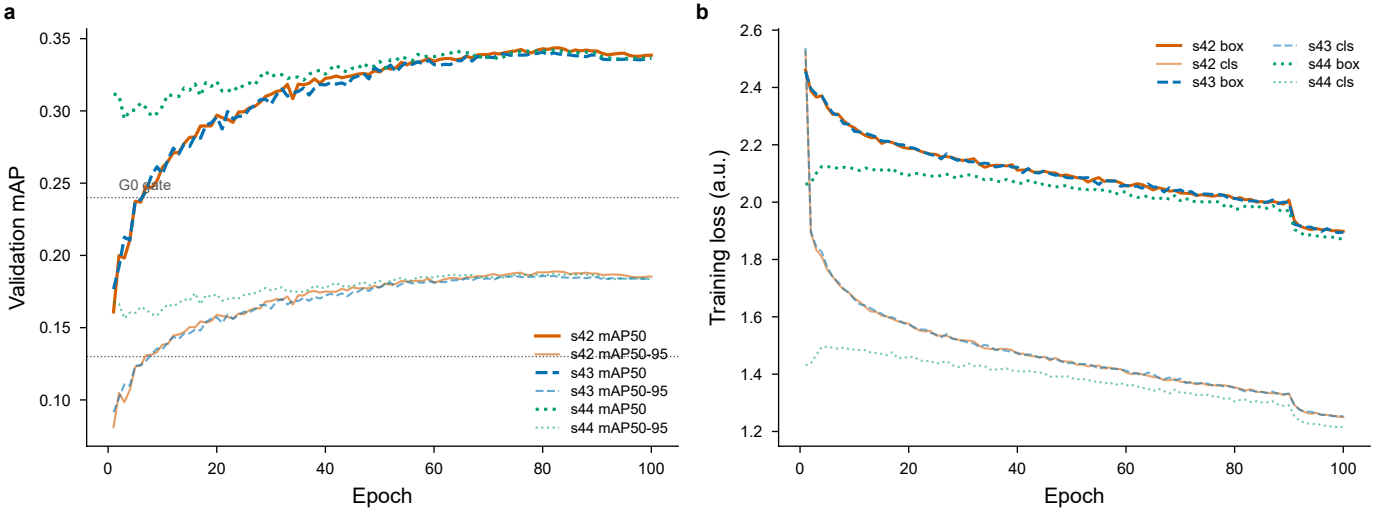
B	util	spend / S	Δ worst o2m	Δ worst o2o	Δ micro o2m	Δ micro o2o
2	0.998	1.00000	−0.63 / −0.63 / −0.63	−0.63 / −0.63 / −0.56	−0.01 / +0.00 / −0.01	+0.01 / +0.01 / +0.01
6	0.990	1.00000	−1.89 / −1.89 / −1.89	+3.99 / +3.66 / −1.89	+0.10 / +0.08 / +0.09	+0.14 / +0.15 / +0.12
10	0.980	1.00000	−2.84 / −2.84 / −3.15	+4.30 / +6.11 / −3.15	+0.31 / +0.32 / +0.27	+0.34 / +0.33 / +0.31
15	0.963	1.00000	+6.38 / +6.38 / +9.40	+6.38 / +6.21 / +6.06	+0.50 / +0.47 / +0.44	+0.60 / +0.56 / +0.54
20	0.945	1.00000	+4.80 / +4.80 / +8.29	+4.80 / +7.98 / +7.98	+0.86 / +0.70 / +0.75	+0.81 / +0.79 / +0.85
25	0.929	1.00000	+11.15 / +8.78 / +11.48	+8.24 / +6.40 / +6.40	+1.00 / +0.96 / +1.08	+1.11 / +1.10 / +1.07
30	0.914	1.00000	+9.58 / +6.89 / +10.22	+7.52 / +4.82 / +4.51	+1.11 / +1.07 / +1.09	+1.21 / +1.11 / +1.19
40	0.882	1.00000	+6.12 / +8.03 / +6.12	+8.42 / +6.43 / +11.44	+1.23 / +1.10 / +1.11	+1.31 / +1.22 / +1.18
50	0.845	1.00000	+5.04 / +6.37 / +2.02	+3.93 / +7.65 / +8.69	+1.00 / +1.17 / +1.13	+1.17 / +1.11 / +1.00
70	0.762	1.00000	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00	+0.01 / +0.02 / +0.01	+0.02 / +0.00 / +0.02
100	0.630	0.70728	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00
300	0.236	0.23576	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00	+0.00 / +0.00 / +0.00

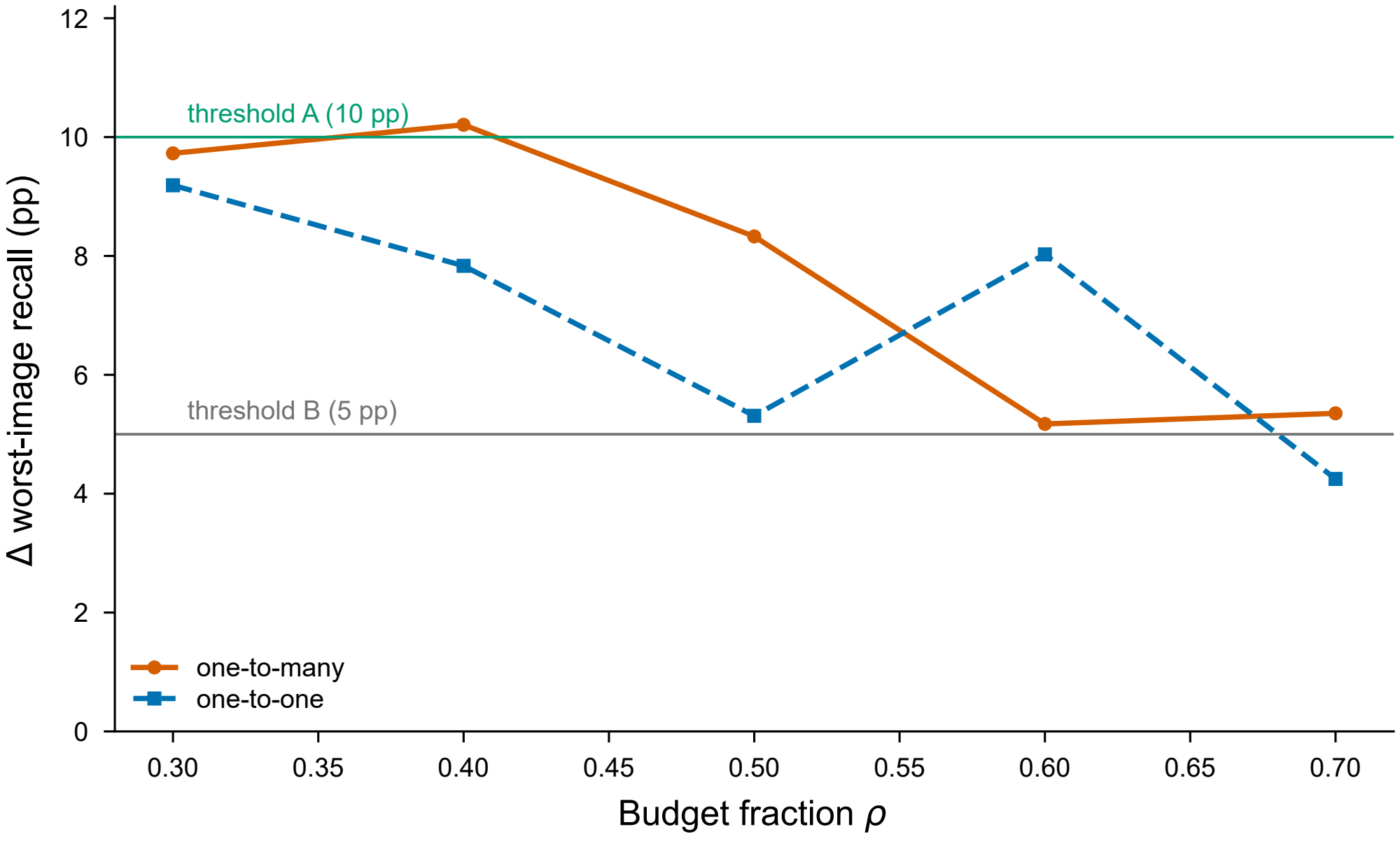
Table S10. Structural audit of the budget sweep, on all three seeds and both heads

The measured gain itself is in Table S9; this table reports when the comparison is well-posed. identical vectors marks budgets at which the uniform and proportional quota vectors are equal entry by entry, so the comparison is an identity rather than a measurement. worst image at cap marks budgets at which the same image is the worst under both allocations. degenerate marks budgets at which some image's quota falls to one box, so its recall is decided by whether the single top prediction is correct.

Values are exactly as computed; see Methods for definitions.

B	identical vectors	worst image at cap (uniform / prop.)	degenerate
2	no	no / no	yes
6	no	no / no	yes
10	no	no / no	yes
15	no	no / no	yes
20	no	no / no	yes
25	no	no / no	yes
30	no	no / no	yes
40	no	no / no	no
50	no	no / no	no
70	no	yes / yes	no
100	yes	yes / yes	no
300	yes	yes / yes	no


a
Validation mAP
0.35
0.30
0.25
0.20
0.15
0.10
G0 gate
s42 mAP50
s42 mAP50-95
s43 mAP50
s43 mAP50-95
s44 mAP50
s44 mAP50-95
0
20
40
60
80
100
Epoch
b
Training loss (a.u.)
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
s42 box
s42 cls
s43 box
s43 cls
s44 box
s44 cls
0
20
40
60
80
100
Epoch


12
10
8
6
4
2
0
threshold A (10 pp)
threshold B (5 pp)
Δ worst-image recall (pp)
one-to-many
one-to-one
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
Budget fraction ρ

Note S1. Numerical self-checks and independent recomputation.

1. Allocation equivalence. Truncating a single inference pass performed at a large max_det reproduces running at each smaller max_det directly: 16 of 16 comparisons match, on both prediction heads and across eight budgets, with a maximum coordinate deviation of 0.0. All budget results therefore cost no additional inference beyond one pass per head.

2. Equal-spend precondition. The two allocations are compared only where their total spends agree to within 1%. Conditions failing this test are reported as excluded rather than silently included, and the count of excluded conditions is stated in the main text.

3. Degeneracy flag. The criterion reports, for every budget, how many images receive a quota of zero or exactly one. In that regime an image's recall is decided by whether the detector's single highest-confidence prediction is correct, so those budgets are excluded from mechanism inference in both directions.

4. Independent recomputation. Chunked evaluation passes are merged only after the merged per-image object counts are re-derived from the label files and compared against the recorded counts; any mismatch aborts the merge. The reported merge validated 548 of 548 images with zero missing and zero mismatches.

5. Two independent paths. Every statistic that entered the pre-registered decision was computed by two separate code paths and required to agree before being reported.